

Constructing an Instrument as a Function of Covariates

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Abstract

Researchers often use instrumental variables (IV) models to investigate the causal relationship between an endogenous variable and an outcome while controlling for covariates. When an exogenous variable is unavailable to serve as the instrument for an endogenous treatment, a recurring empirical practice is to construct one from a nonlinear transformation of the covariates. We investigate how reliable these estimates are under mild forms of misspecification. Our main result shows that for instruments constructed from covariates, the IV estimand can be arbitrarily biased under mild forms of misspecification, even when imposing constant linear treatment effects. We perform a semi-synthetic exercise by calibrating data to alternative models proposed in the literature and estimating the average treatment effect. Our results show that IV specifications that use instruments constructed from covariates are non-robust to nonlinearity in the true structural function.

1 Introduction

Instrumental variable (IV) strategies are widely used to investigate the causal effect of an endogenous variable on an outcome of interest. Since the work of [Imbens and Angrist \(1994\)](#), these estimators are often interpreted as estimating a local average treatment effect (LATE) or, under homogeneity conditions, the population treatment effect of the endogenous variable of interest.

However, estimating this causal effect requires finding valid instruments that satisfy the exclusion restriction, which can be challenging in many applications. One response to this challenge is to construct an instrumental variable that is a nonlinear function of covariates.¹ The leading case is to use the “product instrument,” the interaction of two covariates. A sample of recent articles published in the *American Economic Review* that use this approach are [Munshi and Rosenzweig \(2016\)](#), [Bettinger et al. \(2017\)](#), [Aladangady \(2017\)](#), and [Rogall \(2021\)](#). Other examples include [Shao \(2014\)](#), [Bharadwaj \(2015\)](#), [Helmert et al. \(2017\)](#), [Crawford and Simpson \(2020\)](#), [Cagé et al. \(2022\)](#), [Liu et al. \(2023\)](#), and [Ghose et al. \(2024\)](#).

In this paper, we study the causal interpretability of these IV estimators that use a nonlinear function of covariates as the instrumental variable, under misspecification. A researcher may be interested in the effect of a variable D on some outcome Y , where D may be endogenous to unobserved factors affecting Y . The researcher’s model specifies Y as a linear function of D and some covariates X , with the causal relationships governed by a parameter $\tilde{\theta}$. They do not have access to an excluded variable Z that is correlated with the endogenous variable D but not the unobserved factors affecting the relationship

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¹Here we use covariate to mean “included exogenous variable,” and reserve “excluded variable” to refer to excluded exogenous variables.

between D and Y (conditional upon X). Instead, a recurring practice is for the researcher to construct an instrumental variable $\tilde{f}(X) = X_1 \cdot X_2$ that is the interaction of covariates, which they argue satisfy the IV assumptions. The researcher then uses IV estimators to evaluate $\tilde{\theta}$.

For example, [Aladangady \(2017\)](#) uses a linear model to examine the effect of log house price fluctuations D on log consumer spending Y in the United States while trying to control for potential unobserved confounding caused by preferences or expectations that change over time. They would like to use a measurement of local land availability or zoning regulations Z that is uncorrelated with preference shifts but varies geographically. However, they do not have access to detailed zoning data across regions. Therefore, they use the interaction between long-term real interest rates X_1 and housing supply elasticity measures X_2 as an instrument for house price fluctuations and estimate the two-stage least squares (2SLS) model,

$$Y = \tilde{\pi}_0 + \tilde{\theta}D + \tilde{\pi}_1X_1 + \tilde{\pi}_2X_2 + \tilde{\epsilon}_i, \quad D = \tilde{\tau}_0 + \tilde{\delta}\tilde{f}(X) + \tilde{\tau}_1X_1 + \tilde{\tau}_2X_2 + \tilde{v}_i,$$

for $\tilde{f}(X) = X_1 \cdot X_2$. By including long-term real interest rates X_1 and housing supply elasticity measures X_2 in the specification for consumer spending Y , they hope to control for the *direct effect* of interest rates and housing elasticity on spending trends that may be related to housing prices. This relaxes the exogeneity assumption required on the instrument $\tilde{f}(X)$ because even if long-term interest rates or housing elasticity are correlated with unobserved determinants of spending, as long as their interaction is uncorrelated with these unobserved determinants, the model assumptions required for IV are still satisfied.

We show that for IV estimators that use a function of covariates as an instrument, even while imposing homogeneous treatment effects, the researcher’s model is uninterpretable as a causal effect under mild forms of misspecification. As the main contribution we show that for any level of bias and for any IV estimand which relies on an instrument that is a transformation of covariates $f(X, Z) = \tilde{f}(X)$, there is a continuous data generating process (DGP) with constant, linear treatment effects under which we can attain that level of bias.

Quantitatively, we investigate the sensitivity of linear IV estimates that use a function of covariates $\tilde{f}(X)$ as an instrument to a reasonable set of nonlinear DGPs. For a collection of influential papers that use linear IV with product instruments, we perform a semi-synthetic exercise by estimating the treatment effect of the endogenous variable D under nonlinear structural functions proposed in the literature. For three out of four pairs of papers we examine, we find that a 95% confidence interval for the treatment effect $\hat{\theta}$ rejects the ground truth in our simulations. These findings emphasize the sensitivity of IV estimands that use product instruments to the true relationship between the outcome Y and covariates X in the true structural function. This paper aims to guide the causal interpretation of these estimands.

A large literature following [Imbens and Angrist \(1994\)](#) and [Angrist et al. \(1996\)](#) studies the interpretation of IV estimators under misspecification. Our work is closest to [Blandhol et al. \(2022\)](#), which investigates the LATE interpretation of IV estimators that include exogenous covariates X . Unlike [Blandhol et al. \(2022\)](#), we allow for continuous outcomes and continuous treatment effects. [Blandhol et al. \(2022\)](#) shows that IV estimators with instruments that do not satisfy a condition called “saturated covariates” cannot be interpreted as LATE. As we show in section 2, the estimators we study are undefined under saturated covariates since there is no variation in the instrument $\tilde{f}(X)$. Appendix C expands on

the relationship between our work and [Blandhol et al. \(2022\)](#) and generalizes some of the results in the latter.

Our work is also closely related to [Andrews et al. \(2023\)](#), which studies the causal consistency of nonlinear IV estimators under misspecification. They show that IV estimators with instruments that do not satisfy a condition known as “strong exclusion” cannot approximately describe the effects of the endogenous variable D . We differ in showing a more significant failure of the causal interpretability of IV estimators for instruments of covariates even when imposing homogeneous, linear treatment effects.

Researchers often justify their models using economic reasoning, which can lead them to believe their model is correctly specified. This motivates [Gao and Wang \(2023\)](#), who introduce a causal estimator that does not depend on excluded covariates but is sensitive to the functional form chosen by the analyst. If the researcher observes the true structural function $Y(D, X, \epsilon)$, or if the model proposed by the researcher $\tilde{Y}(D, X, \tilde{\epsilon}_i)$ matches the true DGP, then the estimator from [Gao and Wang \(2023\)](#) and the linear IV specifications we study both remain unbiased. Our work focuses on cases where the researcher does not know the true structural function, so their model may be mildly misspecified.

The remainder of the paper proceeds as follows. Section 2 introduces the assumptions of our model, which includes the IV conditions assumed by the analyst and the true structural function. Section 3 establishes our main result and shows that IV estimators that rely on instruments constructed from covariates can be arbitrarily biased. Finally, Section 4 shows our quantitative results and illustrates the wide range of estimates produced by IV estimators that rely on product instruments across plausible DGPs. We reserve our theoretical proofs for the appendix, with the main text focusing on the key results.

2 Structural model

A analyst observes variables (Y_i, D_i, X_i, Z_i) for units $i \in \{1, \dots, n\}$. All variables are finite-dimensional, and $Y_i \in \mathbb{R}$. To illustrate the possibility of misspecification, we introduce a *true model* that is consistent with the data generating process (DGP) and summarizes the causal relationships between the observed variables, and the *analyst’s* model which may rule out the true DGP.

2.1 True model

Under the true model, which is consistent with the true data-generating process, the observed outcome satisfies $Y_i = Y(D_i, X_i, \epsilon_i)$ for a potential outcome function $d \mapsto Y(d, X_i, \epsilon_i) \in \mathcal{Y} \subseteq \mathbb{R}$, covariates $X_i = (X_{i,1}, \dots, X_{i,J})^\top \in \mathcal{X} \subseteq \mathbb{R}^{\dim(\mathcal{X}) \times 1}$, and unobserved heterogeneity ϵ_i . The observed endogenous variable D_i satisfies $D_i = D(X_i, Z_i, \nu_i)$ for $D(X_i, Z_i, \nu_i) \in \mathcal{D} \subseteq \mathbb{R}$, a potential endogenous variable function, unobserved heterogeneity ν_i , and an excluded exogenous variable $Z_i \in \mathcal{Z} \subseteq \mathbb{R}^{\dim(\mathcal{Z}) \times 1}$. We impose one additional assumption on the true model in line with instrumental variables (IV) models.

Assumption 1 (Excludability). *The potential outcome $Y(\cdot)$ is independent of the excluded exogenous variable Z ,*

$$Y_i = Y(D_i, X_i, Z_i, \epsilon_i) = Y(D_i, X_i, \epsilon_i)$$

Under Assumption 1, we can write $Y_i = Y(D_i, X_i, \epsilon_i)$.

Definition 1 (Linear IV Estimation). Let $f(X, Z)$ be a choice of instrumental variable for $f : \mathcal{X} \times \mathcal{Z} \mapsto \mathbb{R}$. Then, the two-stage least squares (2SLS) IV estimand θ_{IV} is given by,

$$\theta_{IV} = \frac{\mathbb{E}\{Y f_{\perp}\}}{\mathbb{E}\{D f_{\perp}\}} = \frac{\text{Cov}(Y, f_{\perp})}{\text{Cov}(D, f_{\perp})},$$

where $f_{\perp}$ are the residuals from a projection of the instrumental variable $f(X, Z)$ onto X ,

$$f_{\perp} = f(X, Z) - X^{\top} \mathbb{E}\{X X^{\top}\}^{-1} \mathbb{E}\{X f(X, Z)\}.$$

Definition 1 establishes the linear IV estimand to estimate the causal effect of the treatment D on the outcome $Y(D_i, X_i, \epsilon_i)$ under a linear model. This is the estimand computed by linear method of moments and two-stage least squares estimators for an instrumental variable $f(X, Z)$ when there is a single endogenous treatment D .

2.2 Analyst's model

The analyst's model is a special case of the true model. Under the analyst's model, the causal effects of the endogenous variable and included variables on the outcome are governed by a parameter $\tilde{\theta} \in \mathbb{R}$. Here, we focus on the case where the analyst proposes and estimates a linear model with homogeneous treatment effects,

$$Y_i = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) = \tilde{\alpha} + \tilde{\theta} D_i + \tilde{\pi}^{\top} X_i + \tilde{\epsilon}_i. \quad (1)$$

The analyst would like to estimate the causal effect of D on the outcome Y in Equation 1, but is unwilling to assume unconfoundedness ($D \perp\!\!\!\perp \tilde{\epsilon} \mid X$). Therefore, to use IV estimation, the analyst constructs an instrument $f(X, Z)$ under two assumptions sufficient to estimate $\tilde{\theta}$.

Assumption 2 (Exogeneity). *The unobserved heterogeneity ($\tilde{\epsilon}_i$) in Y_i is uncorrelated ($\perp$), or orthogonal, to the instrumental variable $f(X_i, Z_i)$ and covariates X_i ,*

$$f(X, Z), X \perp \tilde{\epsilon}.$$

Assumption 2 says that the instrument $f(X, Z)$ and included exogenous covariates X must be uncorrelated ($\perp$) with the unobserved heterogeneity $\tilde{\epsilon}$ in the outcome under their proposed model.

Assumption 3 (Relevance). *The instrument $f(X, Z)$ is a random variable such that $\mathbb{E}\{D \mid f(X, Z) = w\}$ is a non-trivial function of w , so $\text{Cov}(D, f(X, Z)) \neq 0$.*

Assumption 3 is necessary for θ_{IV} in Definition 1 to be defined, so the denominator is not equal to zero. Otherwise, the treatment effect $\tilde{\theta}$ is unidentified using linear IV estimation.

In this paper, we focus on the case where the analyst does not have access to an excluded exogenous variable Z to create an instrumental variable $f(X, Z)$ that is a nontrivial function of Z . Therefore, the analyst chooses $f(X, Z)$ to be a nonlinear function of only covariates $f(X, Z) = \tilde{f}(X) \in \mathbb{R}$ to use IV estimation.

Lemma 1 (Unsaturated Covariates). *For any function $f(X, Z) = \tilde{f}(X)$ only of covariates, $\text{Var}(f_\perp) \neq 0$ if and only if $\tilde{f}(X)$ is not linear in X for $f_\perp = \tilde{f}(X) - X^\top \mathbb{E}\{XX^\top\}^{-1} \mathbb{E}\{Xf(X, Z)\}$.*

Lemma 1 makes it clear why the analyst’s instrumental variable $\tilde{f}(X)$ must be nonlinear in X to use the IV estimator from Definition 1. If $\tilde{f}(X)$ is linear in each element of X , then the linear IV estimand θ_{IV} becomes unidentified because the denominator $\text{Cov}(D, f_\perp) = \text{Cov}(D, 0) = 0$.

Lemma 2 (Trivial Exogeneity). *Suppose the analyst believes a stronger form of conditional exogeneity, $\mathbb{E}\{\tilde{\epsilon} \mid X\} = 0$. Then, any function of covariates $\tilde{f}(X)$ satisfies exogeneity,*

$$\tilde{f}(X) \perp \tilde{\epsilon}.$$

Lemma 2 demonstrates that an instrument based solely on included exogenous covariates automatically satisfies Assumption 2 (exogeneity) for IV estimation when the analyst assumes a stronger exogeneity condition for these covariates. Some researchers using instruments of the form $f(X, Z) = \tilde{f}(X)$ argue that such instruments “require weaker identifying assumptions than using $[X_1 \text{ or } X_2]$ alone” [Bettinger et al. \(2017\)](#). By including X_i in the second-stage regression, they “control for the direct effect of $[X_i]$ that may be related to $[Y_i]$ ” [Aladangady \(2017\)](#). If the covariates X included in the second-stage equation $\tilde{Y}(D_i, X_i, \tilde{\epsilon}_i)$ satisfy conditional exogeneity ($\mathbb{E}[\tilde{\epsilon} \mid X] = 0$), then a nonlinear function of X can be used as an instrument without needing to independently verify its exogeneity under Assumption 2.

The following proposition shows that this estimand corresponds to the causal effect $\tilde{\theta}$ of D_i on $Y(D_i, X_i, \epsilon_i)$ under the analyst’s model if the true data generating process follows the model assumed by the analyst.

Claim 1 (IV Validity). *Assume the true data generating process $Y(D_i, X_i, \epsilon_i)$ follows the analyst’s constant linear effects model in Equation 1 and Assumption 2,*

$$Y_i = Y(D_i, X_i, \epsilon_i) = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i).$$

and the analyst uses an instrument $f(X, Z) = \tilde{f}(X)$ that is a nonlinear function of covariates. Then, under Assumption 3 (relevance), the IV estimand is equal to the causal effect of D_i on $Y(D_i, X_i, \epsilon_i)$,

$$\theta_{IV} = \frac{\text{Cov}(Y, f_\perp)}{\text{Cov}(D, f_\perp)} = \tilde{\theta},$$

for $f_\perp = \tilde{f}(X) - X^\top \mathbb{E}\{XX^\top\}^{-1} \mathbb{E}\{Xf(X, Z)\}$.

Claim 1 shows that if the analyst’s model is correctly specified and their constructed instrument $f(X, Z) = \tilde{f}(X)$ satisfies Assumption 3 (relevance), IV estimators θ_{IV} ² are consistent for the treatment effect. [Gao and Wang \(2023\)](#) show that even for nonlinear models, if the analyst’s model matches the true DGP, the causal effect θ of D on Y can be recovered locally with estimators of the form introduced in Appendix B. We next move to analyze what happens when the analyst’s model is not correctly specified.

²Note that θ_{IV} is implicitly indexed by $\tilde{f}(X)$

3 Main result: Non-interpretability

When the analyst estimates a linear model like Equation 1, any nonlinear effect of the covariates X on the outcome Y is captured in the unobserved heterogeneity $\tilde{\epsilon}$. This means that IV estimators that rely on moment conditions interacting $\tilde{\epsilon}$ and the covariates X will be sensitive to nonlinearity in the true structural function $Y(D_i, X_i, \epsilon_i)$.

In Theorem 1, we illustrate this sensitivity using a toy case where the true DGP imposes linear, homogeneous treatment effects. We show that if the analyst uses an instrumental variable $f(X, Z) = \tilde{f}(X)$ that is only a function of covariates, the bias of the IV estimand θ_{IV} can be arbitrarily bad.

Theorem 1 (Model bias). *Assume the analyst chooses an instrument $f(X_i, Z_i) = \tilde{f}(X_i)$ that is a non-linear function of covariates.³ If Assumption 3 holds, then, for any true effect $\theta \in \mathbb{R}$, and any level of potential bias $\rho \in \mathbb{R}$, there exists some continuous function $h : X \mapsto \mathbb{R}$ such that if the true DGP satisfies,*

$$Y_i = Y(D_i, X_i, \epsilon_i) = \alpha + \theta D_i + h(X_i) + \epsilon_i, \quad (\epsilon_i \perp\!\!\!\perp D_i, X_i).$$

Then, the bias of the linear IV estimand for the treatment effect is given by $\theta - \theta_{IV} = \rho$.

The proof of Theorem 1 is constructive – we propose an adversarial $h(\cdot)$ which has a nonlinear term proportional to ρ . As we raise ρ , increasing the nonlinearity of the function $h(\cdot)$, the bias of the IV estimand θ_{IV} for the true treatment effect θ increases. This demonstrates the close relationship between the nonlinearity of the true structural function $Y(D_i, X_i, \epsilon_i)$ in X_i and the bias of the linear IV estimand.

Corollary 1 (Product instrument). *Let Assumption 3 hold, and assume the analyst uses an instrument $f(X, Z) = X_1 \cdot X_2$ that is a product of covariates (X_1, X_2) to estimate the IV model in Equation 1. If the true DGP satisfies,*

$$Y_i = Y(D_i, X_i, \epsilon_i) = \alpha + \theta D_i + h(X_i) + \epsilon_i, \quad (\epsilon_i \perp\!\!\!\perp D_i, X_i).$$

Then for,

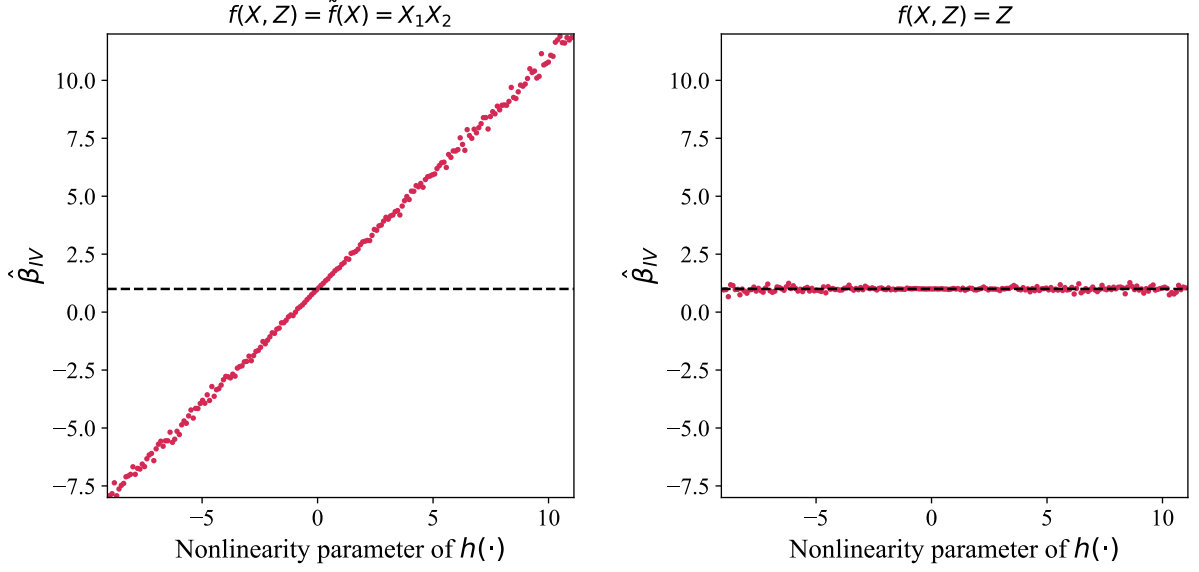
$$h(X_i) = \pi_1 X_{i,1} + \pi_2 X_{i,2} + \rho \frac{\mathbb{E}\{D(X_1 X_2 - X^\top \mathbb{E}\{XX^\top\})\mathbb{E}\{X(X_1 \cdot X_2)\}\}}{\text{Var}(X_1 X_2 - X^\top \mathbb{E}\{XX^\top\})} X_{i,1} X_{i,2},$$

the bias of θ_{IV} for the treatment effect is given by $\theta - \theta_{IV} = \rho$

For the product instrument $\tilde{f}(X) = X_1 \cdot X_2$, which is the instrument used amongst the papers cited in the introduction, Corollary 1 gives the explicit form of the adversarial $h(\cdot)$ from Theorem 1. This function $h(\cdot)$ is constructed from a linear term and a nonlinear term proportional to $X_1 \cdot X_2$. As we enlarge the nonlinear term, the bias of the IV estimand θ_{IV} increases. To illustrate this effect graphically, figure 1 plots IV estimates $\hat{\beta}_{IV}$ against data generated using this function to show the non-interpretability of the IV estimand under misspecification.

³Recall from Lemma 2, that this is necessary for θ_{IV} to be well-defined.

Figure 1: Simulation of IV estimates for the DGP $Y_i = \alpha + \theta D_i + h(X_i) + \epsilon_i$.



(a) Instrumental variable that is nonlinear function of included variables (X_1, X_2) (b) Instrumental variable that is function of excluded variable Z

Notes: Each dot represents an estimate of θ_{IV} from Definition 1 using 2SLS with 1,000 generated observations. The dashed line indicates the true effect $\theta = 1$ of D on Y in the simulations.

Examining panel (a) of Figure 1 illustrates the sensitivity of these IV estimates to misspecification in the true relationship between X and Y . As Corollary 1 implies, we see that by increasing the nonlinearity in $h(\cdot)$ we can inflate the bias of the IV estimate $\hat{\theta}$ while holding the true treatment effect θ constant. In contrast, the consistency of the IV estimates in panel (b) underscores the robustness of the IV estimator against misspecification in the true relationship between X and Y for partially linear DGPs when using a product instrument.

Remark 1 (OVB). The bias from Theorem 1 can be interpreted as a form of omitted variable bias (OVB), resulting from the exclusion of the nonlinear effect of the covariates X on the outcome Y from the analyst's model.

Remark 2 (Rich models). Theorem 1 shows that the linear IV estimand can have arbitrary bias even for simple DGPs with homogeneous linear effects. The proof further implies that estimates can also be biased for richer structural functions $Y(X_i, D_i, \epsilon_i)$ that are nonlinear in X .

Remark 1 gives an alternative interpretation of the bias for IV estimators. Remark 2 underlines the generality of Theorem 1: for IV estimands that rely on instruments of covariates $\tilde{f}(X)$, if the analyst's model does not capture how the covariates X enter the true structural function $Y(D_i, X_i, \epsilon_i)$, then the IV estimand can be very biased.

We have shown that the practice of using instruments constructed from nonlinear functions of covariates $f(X, Z) = \tilde{f}(X)$ can misbehave in interesting ways theoretically. In Section 4, we demonstrate that this behavior is important in empirical work.

4 Application

Researchers often have strong economic reasoning to justify their proposed model and may believe that their model matches the true DGP. In these cases, Claim 1 (IV validity) shows that the IV estimand θ_{IV} gives the average causal effect of D on $Y(D_i, X_i, \epsilon_i)$.

However, it is also often the case in the literature that non-nested models are used to estimate the same treatment effect. In these instances, every model cannot match the true DGP, and therefore some of these models must be misspecified.

For example, recall from Section 1 that Aladangady (2017) uses IV estimation to investigate the effect of log changes in house price D on log consumer expenditure Y in the United States using a model of the form,

$$Y_i = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i,$$

for $X \in \mathbb{R}^2$. Real interest rates X_1 and housing supply elasticity measures X_2 are included as covariates.

Similarly, Disney et al. (2010) estimates the effect of log house price growth D on log consumption Y in the United Kingdom, but includes interest rates X_1 and future expectations X_3 in their life-cycle model for expenditure. They use ordinary least-squares to estimate the model,

$$Y_i = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top \log(X_i) + \tilde{\epsilon}_i,$$

for $X \in \mathbb{R}^7$,⁴ where we write $\log(X)$ to be $\log(X_i) = (\log(X_1), \log(X_2), \dots)$.

Both authors justify their models with economic intuition. Still, they do not agree on the functional form of consumer expenditure, or the way the covariates X enter the model $\tilde{Y}(D_i, X_i, \epsilon_i)$. Theorem 1 (model bias) shows that if the author's model *does not* match the true structural function, IV estimators that use an instrument constructed from covariates can have a high level of bias. Therefore, it is important to understand the sensitivity of these estimates to how the covariates X affect the outcome Y in cases when there are several reasonable choices to model the DGP.

Table 1 gives additional examples of competing models used in the literature. For each of these examples, we compare: on one hand, (i) a linear IV model that relies on the product instrument $\tilde{f}(X) = X_1 \cdot X_2$; and on the other hand (ii) a richer nonlinear model proposed in that literature to estimate the same treatment effect. For each pair of papers, the linear IV model is written on the top row and the non-nested alternative model is given on the bottom row.

⁴The authors also include gross household income, passive savings, estimated unanticipated house growth, and household characteristics as covariates.

Table 1: Alternative choices for $\tilde{Y}(D_i, X_i, \tilde{\epsilon}_i)$

Paper	Model	Outcome Y	Endo. Variable D
Aladangady (2017)	$Y_i = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	Log consumption	Log house wealth
Disney et al. (2010)	$Y_i = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top \log(X_i) + \tilde{\epsilon}_i$	Log consumption	Log house wealth
Munshi and Rosenzweig (2016)	$Y_i = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	Migration indicator	Level of wealth
Abramitzky (2008)	$Y_i = \Phi(\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i) + \tilde{\epsilon}_i$	Migration indicator	Level of wealth
Helmert et al. (2017)	$Y_i = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	Patent count	Network size
Chang et al. (2006)	$Y = \exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\} + \tilde{\epsilon}_i$	Patent count	Network size
Liu et al. (2023)	$Y_i = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	Indicator of child	Value of home
Clark and Ferrer (2019)	$Y_i = \frac{\exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\}}{1 + \exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\}} + \tilde{\epsilon}_i$	Indicator of child	Value of home

Notes: Of the papers cited in the introduction that rely on IV estimation using the product instrument $\tilde{f}(X) = X_1 \cdot X_2$, Table 1 gives an overview of the alternative models used among papers cited in their literature review.

Table 1 shows that cases often arise when there is no consensus about the functional form of the true structural function $Y(D_i, X_i, \epsilon_i)$. Therefore, it is likely that the true DGP is nonlinear in the covariates X .

To assess the sensitivity of linear IV models that use product instruments, Table 2 performs a semi-synthetic exercise for each pair of papers in Table 1. Specifically, for each pair of papers, we interpret the richer, nonlinear specification as the true structural equation and calibrate the DGP to match that model. We restrict one degree of freedom for the parameters $(\alpha^\top, \pi^\top)$ by forcing $\theta = \mathbb{E}\left\{\frac{\partial}{\partial \alpha_1} Y(D_i, X_i, \epsilon_i)\right\} = 1$. Otherwise the choice for $(\alpha^\top, \pi^\top)$ is unconstrained. We then use 2SLS to estimate $\hat{\theta}$ in the linear model IV model from Equation 1.

For example, using the structural model offered by [Disney et al. \(2010\)](#) we construct synthetic draws of (Y_i, D_i, X_i) such that $\theta = \alpha_1 = 1$. Then, we fit the linear IV model used by [Aladangady \(2017\)](#) and compare how the IV estimate $\hat{\theta}$ compares to θ in the synthetic structural model.

4.1 Results

Table 2 shows estimates of $\hat{\theta}$ from Equation 1 for the estimand θ_{IV} in Definition 1 using linear IV estimation.

Table 2: Estimates of $\hat{\theta}$ from Equation 1 using 2SLS

Paper	Model: $\tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) =$	DGP: $Y(D_i, X_i, \epsilon_i) =$	$\hat{\theta}$
Aladangady (2017)	$\tilde{\alpha} + \tilde{\theta}D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	$\alpha_0 + \alpha_1 D_i + \pi^\top \log(X_i) + \epsilon_i$	1.38 (1.01, 1.74)
Munshi and Rosenzweig (2016)	$\tilde{\alpha} + \tilde{\theta}D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	$\Phi(\alpha_0 + \alpha_1 D_i + \pi^\top X_i) + \epsilon_i$	0.55 (0.46, 0.65)
Helmert et al. (2017)	$\tilde{\alpha} + \tilde{\theta}D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	$\exp\{\alpha_0 + \alpha_1 D_i + \pi^\top X_i\} + \epsilon_i$	1.45 (0.85, 2.05)
Liu et al. (2023)	$\tilde{\alpha} + \tilde{\theta}D_i + \tilde{\pi}^\top X_i + \tilde{\epsilon}_i$	$\frac{\exp\{\alpha_0 + \alpha_1 D_i + \pi^\top X_i\}}{1 + \exp\{\alpha_0 + \alpha_1 D_i + \pi^\top X_i\}} + \epsilon_i$	0.48 (0.39, 0.58)

Notes: Each row estimates $\hat{\theta}$ from Equation 1 using the IV model from the corresponding paper. Data is generated following the richer, nonlinear model, i.e. the structural function listed under DGP. $\hat{\theta}$ reports the average estimate across 500 simulations for the estimand θ_{IV} in Definition 1. 95% confidence intervals for $\hat{\theta}$ across simulations are reported in parenthesis. For each DGP, $(\alpha^\top, \pi^\top)$ are chosen while maintaining the condition $\theta = \mathbb{E}\left\{\frac{\partial}{\partial \alpha_1} Y(D_i, X_i, \epsilon_i)\right\} = 1$. Details of this calculation are at the end of Appendix A.

In Table 2, we observe that for three out of four pairs of papers, a 95% confidence interval for the estimated treatment effect $\hat{\theta}$ does not include the ground truth of $\theta = 1$, i.e. the average partial treatment effect (APE) in the true structural function $Y(D_i, X_i, \epsilon_i)$. For instance, IV estimates tend to overestimate the true APE when using the model from [Aladangady \(2017\)](#) where the DGP is partially linear with log-transformed covariates X . Meanwhile, for the models used by [Munshi and Rosenzweig \(2016\)](#) and [Liu et al. \(2023\)](#), where the true structural function follows a probit and logit design, IV estimates based on instruments that are functions of covariates tend to underestimate the APE. This illustrates how the bias of the IV estimand can be very positive or negative when using product instruments depending on the underlying DGP.

These findings underscore the sensitivity of these IV estimates $\hat{\theta}$ to the functional form chosen to model the relationship between covariates X and the outcome Y and illustrate their bias under sensible choices for the DGP. Even for reasonable choices of $Y(D_i, X_i, \epsilon_i)$, IV estimates of the treatment effect $\hat{\theta}$ with constructed instruments can differ substantially from the true APE.

Another question we can ask is how estimates change if we fit different kinds of models to real economic data. To demonstrate the sensitivity of causal conclusions to the functional form of the analyst’s model using real data, Appendix B estimates θ using data from the subset of papers published in the *American Economics Review* (AER) since 2014 which use IV specifications that rely on product instruments. We choose different analyst’s models from Table 1, and fit these models to the data. Together, these findings give empirical evidence that IV specifications using product instruments are non-robust to nonlinearity in the true structural function.

Overall, our results underscore the importance of functional form choices in IV models where instruments are functions of covariates, $f(X, Z) = \tilde{f}(X)$. The selected functional form can significantly influence how the treatment effect is interpreted.

5 Recommendations for practice

In this paper, we have provided theoretical and empirical evidence showing that product instruments, or more generally, instruments that are transformations of covariates, can be very biased. Our framework shows that even when imposing homogeneous linear effects between an endogenous variable and outcome, these IV estimates cannot be interpreted as a causal effect under mild forms of misspecification. Theoretically, we show that there exists a simple DGP with constant linear effects that can lead to unbounded bias of the IV estimand. Empirically, we illustrate the sensitivity of IV estimates that use instruments constructed from nonlinear functions of covariates.

When a researcher does not have access to an excluded, exogenous variable, we recommend caution when discussing the causal interpretability of IV estimators. Alternatively, if the researcher believes their instrumental variable $f(X, Z) = \tilde{f}(X)$ satisfies the IV assumptions in Section 2, it is important that they strongly defend their choice of model $\tilde{Y}(\cdot)$, i.e. the relationship between the outcome Y and covariates X , to defend the validity of their estimate.

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Appendix for “Constructing Instruments as a Function of Included Covariates”

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A Proofs from main text

Proof of Lemma 1.

To prove the forward direction, let $\text{Var}(f_\perp) = \mathbb{E}\{f_\perp^2\} = 0$. This implies that $f_\perp = 0$ almost surely. However, since $X^\top \mathbb{E}\{XX^\top\}^{-1} \mathbb{E}\{X\tilde{f}(X)\}$ is a linear projection of $\tilde{f}(X)$ onto X , this implies that $\tilde{f}(X)$ is a linear function of each $X_{,j} \in X$.

Similarly to prove the backwards direction, let $\text{Var}(f_\perp) = \mathbb{E}\{f_\perp^2\} \neq 0$. Since $X^\top \mathbb{E}\{XX^\top\}^{-1} \mathbb{E}\{X\tilde{f}(X)\}$ is a linear projection of $\tilde{f}(X)$ onto X , this implies that $f_\perp$ is not equal to 0 almost surely, so it is not a linear function of each $X_{,j} \in X$. The result follows. $\square$

Proof of Lemma 2.

Using the tower property, notice that,

$$\text{Cov}(\tilde{f}(X_i), \tilde{\epsilon}_i) = \mathbb{E}\{\tilde{\epsilon}_i \tilde{f}(X_i)\} - \mathbb{E}\{\tilde{\epsilon}_i\} \mathbb{E}\{\tilde{f}(X_i)\} = \mathbb{E}\{\mathbb{E}\{\tilde{\epsilon}_i \mid X_i\} \tilde{f}(X_i)\} = 0.$$

$\square$

Proof of Claim 1.

We assume that the true DGP matches the analyst’s model, so,

$$Y_i = Y(D_i, X_i, \epsilon_i) = \alpha_i + D_i \theta + \pi^\top X_i + \epsilon_i.$$

Then since under Assumption 2, we have that $\text{Cov}(\epsilon_i, f_\perp) = 0$ we can write,

$$\begin{aligned} \theta_{IV} &= \frac{\text{Cov}(Y, f_\perp)}{\text{Cov}(D, f_\perp)} = \frac{\text{Cov}(Y(D, X, \epsilon), f_\perp)}{\text{Cov}(D, f_\perp)} \\ &= \frac{\text{Cov}(\alpha, f_\perp)}{\text{Cov}(D, f_\perp)} + \frac{\text{Cov}(\theta D, f_\perp)}{\text{Cov}(D, f_\perp)} + \frac{\text{Cov}(X\pi, f_\perp)}{\text{Cov}(D, f_\perp)} + \frac{\text{Cov}(\epsilon, f_\perp)}{\text{Cov}(D, f_\perp)} \\ &= \theta \frac{\text{Cov}(D, f_\perp)}{\text{Cov}(D, f_\perp)} + \pi \frac{\text{Cov}(X, f_\perp)}{\text{Cov}(D, f_\perp)} \\ &= \theta. \end{aligned}$$

The assertion that $\text{Cov}(X_i, f_\perp) = 0$ in the last line follows since $f_\perp = \tilde{f}(X) - X^\top \mathbb{E}\{XX^\top\}^{-1} \mathbb{E}\{X\tilde{f}(X)\}$ are the residuals from the projection of $\tilde{f}(X)$ onto X , and $\text{Cov}(D, f_\perp) \neq 0$ follows from Assumption 3 (relevance). $\square$

Proof of Theorem 1.

To prove Theorem 1, we will construct a continuous function $h(X_i)$ such that the estimand $\theta_{IV} = \rho$. Let

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$f_{\perp} = \tilde{f}(X) - X\mathbb{E}\{XX^{\top}\}^{-1}\mathbb{E}\{X\tilde{f}(X)\}$ be the residuals from a projection of the instrumental variable $\tilde{f}(X)$ onto X and $g(\rho) = \frac{\text{Cov}(D, f_{\perp})}{\text{Var}(f_{\perp})}\rho$. Consider the function,

$$Y_i = Y(D_i, X_i, \epsilon_i) = \alpha + \theta D_i + h(X_i) + \epsilon_i$$

$$\text{for } h(X_i) = X_i\pi + g(\rho)\tilde{f}(X_i).$$

Then, we can write the IV estimand as,

$$\begin{aligned} \theta_{IV} &= \frac{\text{Cov}(Y, f_{\perp})}{\text{Cov}(D, f_{\perp})} = \frac{\text{Cov}(\alpha + \theta D_i + h(X_i) + \epsilon_i, f_{\perp})}{\text{Cov}(D, f_{\perp})} \quad (\text{I}) \\ &= \frac{\text{Cov}(\alpha, f_{\perp})}{\text{Cov}(D, f_{\perp})} + \frac{\text{Cov}(\theta D, f_{\perp})}{\text{Cov}(D, f_{\perp})} + \frac{\text{Cov}(X\pi, f_{\perp})}{\text{Cov}(D, f_{\perp})} + \frac{\text{Cov}(g(\rho)\tilde{f}(X_i), f_{\perp})}{\text{Cov}(D, f_{\perp})} + \frac{\text{Cov}(\epsilon, f_{\perp})}{\text{Cov}(D, f_{\perp})} \\ &= \theta \frac{\text{Cov}(D, f_{\perp})}{\text{Cov}(D, f_{\perp})} + \frac{\text{Cov}(X, f_{\perp})}{\text{Cov}(D, f_{\perp})}\pi + g(\rho) \frac{\text{Var}(f_{\perp})}{\text{Cov}(D, f_{\perp})} \quad (\text{II}) \\ &= \theta + g(\rho) \frac{\text{Var}(f_{\perp})}{\text{Cov}(D, f_{\perp})} \quad (\text{III}) \\ &= \rho. \end{aligned}$$

Where (I) follows because we assumed the DGP satisfies $Y = \alpha + \theta D_i + h(X_i) + \epsilon_i$. $\text{Cov}(\epsilon, f_{\perp}) = 0$ in line (II) follows since we impose $\epsilon \perp X$. $\text{Cov}(X, f_{\perp}) = 0$ in line (III) follows since $f_{\perp} = \tilde{f}(X) - X^{\top}\mathbb{E}\{XX^{\top}\}^{-1}\mathbb{E}\{X\tilde{f}(X)\}$ are the residuals from the projection of $\tilde{f}(X)$ onto X , and $\text{Cov}(D, f_{\perp}) \neq 0$ follows from Assumption 3 (relevance). $\square$

Calibration of parameters in Table 2.

Let the richer structural function used as the DGP be denoted by $g(\cdot)$. Then, we are given a nonlinear, continuous function $g(\alpha_1 D_i + \pi^{\top} X_i + \epsilon_i)$ where (D_i, X_i, ϵ_i) are chosen to satisfy,

$$(D_i, \pi^{\top} X_i, \epsilon_i)^{\top} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma) \quad \text{for } \boldsymbol{\mu} = \begin{pmatrix} \frac{\mu_1}{\alpha_1} \\ \mu_2 \\ -\mu_1 - \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma^2 & \rho_1 & 0 \\ \rho_1 & V_2 & 0 \\ 0 & 0 & V_3 \end{pmatrix}$$

Letting $V = V_2 + V_3$, it follows that we can express $g(\alpha_1 D_i + \pi^{\top} X_i + \epsilon_i) \sim g(Y)$ for $Y \sim \mathcal{N}(0, \alpha_1^2(\sigma^2 + 2\rho_1) + V)$. Note that using the chain rule,

$$\frac{\partial}{\partial \alpha_1} g(\alpha_1 D_i + \pi^{\top} X_i + \epsilon_i) = \alpha_1 g'(\alpha_1 D_i + \pi^{\top} X_i + \epsilon_i) \sim \alpha_1 g'(Y)$$

For a given π , it follows that we can choose a solution for α_1 that satisfies $\mathbb{E}\{\alpha_1 g'(Y)\} = 1$ by minimizing $\left(\alpha_1 - \mathbb{E}\{g'(Y)\}^{-1}\right)^2$ for $\alpha_1 > 0$.

$\square$

B Empirical benchmarking

We take the data from a collection of papers published in the *American Economic Review* (AER) since 2014 which use instruments of interacted covariates $f(X, Z) = \tilde{f}(X)$ that we could obtain the replication code for and estimate the treatment effect ($\tilde{\theta}$) under different model specifications. Under the framework of [Gao and Wang \(2023\)](#), still imposing homogenous treatment effects we estimate models of the form,

$$Y_i = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) = h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) + \tilde{\epsilon}_i, \quad (2)$$

For an endogenous treatment D_i , exogenous covariates X_i , and unobserved heterogeneity $\tilde{\epsilon}_i$.

B.1 Results

Because the papers we examine *do not* have access to an excluded exogenous variable Z , we must develop an identification argument for the treatment effect that can be applied for nonlinear models of the form in Equation 2. We present the full argument in Appendix B.2.

For the set of alternative specifications $h(D_i, X_i; \tilde{\alpha}, \tilde{\pi})$ of the relationship between the outcome Y and included covariates X shown in Table 1, we apply this estimator to a subset of papers published in the *American Economic Review* (AER). Table 1 shows the estimates computed for each paper.²

²Paper (1) is [Rogall \(2021\)](#) and paper (2) is [Munshi and Rosenzweig \(2016\)](#).

Appendix Table 1: Estimates of $\tilde{\theta} = \mathbb{E}\left\{\frac{\partial}{\partial D_i} h(D_i, X_i; \tilde{\alpha}, \tilde{\pi})\right\}$ from Equation 2.

$h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) =$	Paper		
	(1)	(2a)	(2b)
$\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i$ (<i>reported</i>)	5.56 (3.60, 7.52)	1.49 (-0.47, 3.45)	-0.70 (-2.66, 1.26)
$\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top \exp\{X_i\}$	3.05 (2.86, 3.26)	0.21 (-0.03, 0.51)	-0.55 (-0.74, -0.36)
$\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top \log(1 + X_i)$	3.33 (3.15, 3.50)	0.21 (-0.28, 0.63)	-0.48 (-0.68, -0.28)
$\exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\} + c$	3.41 (3.20, 3.61)	0.68 (0.35, 1.07)	-0.70 (-0.92, -0.49)
$\log(1 + \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i)$	1.39 (0.63, 2.62)	0.27 (-0.09, 0.71)	-0.43 (-0.65, -0.22)
$k \frac{\exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\}}{1 + \exp\{\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i\}} + c$	8.08 (7.55, 8.49)	0.31 (-0.07, 0.66)	-0.48 (-0.70, -0.28)
$k\Phi(\tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top X_i) + c$	2.23 (2.07, 2.49)	0.38 (0.03, 0.72)	-0.48 (-0.69, -0.29)

Notes: Each row estimates the average partial treatment effect $\tilde{\theta}$ for the specification listed under $h(D_i, X_i; \tilde{\alpha}, \tilde{\pi})$ by minimizing the objective function from Equation 4. We standardized each column by dividing it by the standard deviation of the linear estimate, which corresponds to the IV estimate using the authors' instrument (reported) for papers 1 & 2. Parametric bootstrap 95% confidence intervals following Appendix B.2 with 250 replicates are in parenthesis. (*) For the logit and probit models and papers with non-binary outcomes, we scale the model to cover the support of Y with $k = \text{range}(Y)$ and $c = \min(Y)$.

Across different choices for $h(D_i, X_i; \tilde{\alpha}, \tilde{\pi})$ within the same article, we see that many of the estimates of the causal effect $\tilde{\theta}$ of D_i on Y_i are significantly different from each other. Taking paper (1) as an example, a 95% confidence interval for θ using the estimate $\hat{\theta}$ and standard error of the reported linear specification does not overlap with the 95% confidence interval using the log specification $h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) = \tilde{\alpha}_0 + \tilde{\alpha}_1 D_i + \tilde{\pi}^\top \log(1 + X_i)$.

This illustrates the sensitivity of an analyst's estimate of $\tilde{\theta}$ to their choice of function form to model the relationship between the included covariates X and outcome Y . Even for reasonable choices of $h(D_i, X_i; \tilde{\alpha}, \tilde{\pi})$, the magnitude and significance of the causal effect between D and Y implied by the researcher's estimate can vary considerably.

These results emphasize that under IV models that use instruments which are functions of included covariates $f(X, Z) = \tilde{f}(X)$, or other models that rely on the correct specification between X and Y , the choice of functional form can have a large impact on the interpretation of the treatment effect.

B.2 Estimation

In order to estimate the causal effect of Equation 2 without excluded exogenous covariates Z_i , we need two additional assumptions on $\tilde{Y}(D_i, X_i, \tilde{\epsilon}_i)$.

Assumption 4 (Strong Exogeneity). *The included exogenous covariates X satisfy “strong” exogeneity,*

$$X \perp\!\!\!\perp \tilde{\epsilon}.$$

Under Assumption 4, we can rewrite Equation 2 in terms of only exogenous covariates X_i ,

$$\begin{aligned} \mathbb{E}\{Y_i \mid X_i\} &= \mathbb{E}\{h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) \mid X_i\} + \mathbb{E}\{\tilde{\epsilon}_i \mid X_i\} = \mathbb{E}\{h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) \mid X_i\}, \\ &\Rightarrow u(X_i) = v(X_i; \tilde{\alpha}, \tilde{\pi}). \end{aligned} \quad (3)$$

Where $u(X_i) = \mathbb{E}\{Y_i \mid X_i\}$ and $v(X_i; \tilde{\alpha}, \tilde{\pi}) = \mathbb{E}\{h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) \mid X_i\}$ can both be approximated by non-parametric regressions. When represented as a function *only* of exogenous covariates, under one additional assumption can estimate $(\hat{\alpha}, \hat{\pi})$.

Assumption 5 (Full Rank). *Assume the function $v(X_i; \tilde{\alpha}, \tilde{\pi})$ is continuous in each $X_{i,j} \in (X_{i,1}, \dots, X_{i,J}) \in \mathbb{R}^{1 \times J}$ and $\mathbb{E}\{\nabla_{(\tilde{\alpha}, \tilde{\pi})} v(X_i; \tilde{\alpha}, \tilde{\pi}) \nabla_{(\tilde{\alpha}, \tilde{\pi})}^\top v(X_i; \tilde{\alpha}, \tilde{\pi})\}$ is full rank.*

As we will show, Assumption 5 is necessary to solve an optimization problem of the form,

$$(\hat{\alpha}, \hat{\pi})^\top = \arg \min_{\tilde{\alpha} \in \mathbf{A}, \tilde{\pi} \in \Pi} \frac{1}{n} \sum_{i=1}^n (u(X_i) - v(X_i; \tilde{\alpha}, \tilde{\pi}))^2. \quad (4)$$

Assumptions 4 and 5 together allow us to estimate the parameters in Equation 2 when the model is correctly specified.

Proposition 1. *Assume that the model proposed by the analyst in equation 2 is correctly specified,*

$$Y_i = \tilde{Y}(D_i, X_i, \tilde{\epsilon}_i) = h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) + \tilde{\epsilon}_i.$$

Then, under Assumptions 4 and 5, it follows that the model parameters $(\tilde{\alpha}, \tilde{\pi})$ are locally identified.

Proposition 1 gives us a way to estimate the model parameters $(\tilde{\alpha}, \tilde{\pi})$ under different specifications of $Y_i = \tilde{Y}(D, X, \tilde{\epsilon})$ without the existence of an excluded exogenous instrument Z . Specifically, we,

1. Use kernel ridge regression with $\lambda = 1/\sqrt{n}$ to estimate $\mathbb{E}\{Y \mid X_i = x\} = \hat{u}(x)$
2. For each $(\tilde{\alpha}, \tilde{\pi}) \in \mathbf{A} \times \Pi$,
 - (a) Use kernel ridge regression with $\lambda = 1/\sqrt{n}$ to estimate $\mathbb{E}\{h(D_i, X_i; \tilde{\alpha}, \tilde{\pi}) \mid X_i = x\} = \hat{v}(x; \tilde{\alpha}, \tilde{\pi})$
 - (b) Calculate the sample error, $R(\tilde{\alpha}, \tilde{\pi}) = \frac{1}{n} \sum_{i=1}^n (\hat{u}(X_i) - \hat{v}(X_i; \tilde{\alpha}, \tilde{\pi}))^2$
3. Choose the estimates,

$$(\hat{\alpha}, \hat{\pi}) = \arg \min_{\tilde{\alpha} \in \mathbf{A}, \tilde{\pi} \in \Pi} \frac{1}{n} \sum_{i=1}^n (u(X_i) - v(X_i; \tilde{\alpha}, \tilde{\pi}))^2.$$

Gao and Wang (2023) show that, for $V_0 = \mathbb{E}\{\nabla_{(\tilde{\alpha}, \tilde{\pi})}^\top v(X_i; \tilde{\alpha}, \tilde{\pi}) \nabla_{(\tilde{\alpha}, \tilde{\pi})} v(X_i; \tilde{\alpha}, \tilde{\pi})\}$, under standard regularity conditions the limit distribution for this estimate is given by,

$$\sqrt{n}((\hat{\alpha}, \hat{\pi})^\top - (\tilde{\alpha}, \tilde{\pi})^\top) \xrightarrow{d} \mathcal{N}(0, \Sigma), \quad \Sigma = V_0^{-1} \mathbb{E}\{\tilde{\epsilon}_i^2 \nabla_{(\tilde{\alpha}, \tilde{\pi})}^\top v(X_i; \tilde{\alpha}, \tilde{\pi}) \nabla_{(\tilde{\alpha}, \tilde{\pi})} v(X_i; \tilde{\alpha}, \tilde{\pi})\} V_0^{-1}$$

We then define $\hat{\theta}$ as the average partial treatment effect of the endogenous variable D_i on the outcome Y_i using our estimates,

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial D_i} h(D_i, X_i; \hat{\alpha}, \hat{\pi}).$$

We approximate the standard error of $\hat{\theta}$ by taking bootstrap draws from the limit distribution of $(\hat{\alpha}, \hat{\pi})^\top$ and calculating the average partial treatment effect for each draw.

C Weak Causality

C.1 Nonlinear IV and LATE interpretation

In this section, we generalize the theoretical results in [Blandhol et al. \(2022\)](#) to cover continuous random variables (Y, D, Z, X) and non-linear treatment effects. We then use the results to analyze the local average treatment effects (LATE) interpretation of IV estimators of instruments $f(X, Z) = \tilde{f}(X)$ that are nonlinear functions of exogenous variables.

Definition 2 (Weakly Causal). β is *weakly causal* if both of the following statements are true,

- a If $Y(d, x, \epsilon)$ is non-decreasing in d for all $d \in \mathcal{D}$ and $x \in \mathcal{X}$, then $\beta \geq 0$
- b If $Y(d, x, \epsilon)$ is non-increasing in d for all $d \in \mathcal{D}$ and $x \in \mathcal{X}$, then $\beta \leq 0$

Definition 2 is a minimal requirement for an estimand β to reflect the causal effect of D on Y . It says that if the causal effect of the treatment has the same sign for individual, then the summary estimand β also has that sign. An estimand β which fails to satisfy Definition 2 becomes uninterpretable as a measure of the treatment effect D on the outcome Y . Note that this definition of weak causality based on the potential outcomes implies the definition in [Blandhol et al. \(2022\)](#).

Assumption 6 (Instrument Monotonicity). For the potential endogenous variable function $D_i = D(X_i, Z_i, v_i)$ and instrument $f(X_i, Z_i)$ assume either,

- a $D(x_i, z_i, v_i)$ weakly increases as $f(x_i, z_i)$ increases for all $z \in \mathcal{Z}$ and $x \in \mathcal{X}$
- b $D(x_i, z_i, v_i)$ weakly decreases as $f(x_i, z_i)$ increases for all $z \in \mathcal{Z}$ and $x \in \mathcal{X}$

Assumption 6 requires that there is a consistent relationship between the endogenous variable D and the instrument $f(X, Z)$. It says that as the observed instrument $f(X_i, Z_i)$ increases, so does the endogenous variable D_i . Using assumption 6, we can arrive at proposition 4 of [Blandhol et al. \(2022\)](#) without assuming discrete random variables or constant, linear treatment effects.

Proposition 2 (Weakly Causal). Let Assumptions 1, 2, 3, and 6 hold. Then, θ_{IV} is weakly causal if and only if $\mathbb{E}\{f_\perp \mid X\} = 0$.

Corollary 2 (model bias). For any instrumental variable $f(X, Z) = \tilde{f}(X)$ that is a function exclusively of included covariates X , θ_{IV} is not weakly causal.

Corollary 2 gives an alternate result to Theorem 1 that shows how IV estimands of instrumental variables $\tilde{f}(X)$ that are only functions of included covariates lack causal interpretability. It relies on the

failure of the “saturated covariates” condition ($\mathbb{E}\{f_\perp \mid X\} = 0$) for IV estimands which use an instrument constructed from a nonlinear function of included covariates. Corollary 2 implies that these estimators cannot be interpreted as LATE.

D Proofs from appendix

Proof of Proposition 1.

Notice that if the model in Equation 2 is correctly specified, then $\mathbb{E}\{Y \mid X\} = u(X) = v(X; \tilde{\alpha}, \tilde{\pi})$. Therefore under Assumption 4 (strong exogeneity), for all $x \in \mathcal{X}$ the parameters $(\tilde{\alpha}, \tilde{\pi})$ should satisfy,

$$R(x; \tilde{\theta}, \tilde{\pi}) = u(x) - v(x; \tilde{\alpha}, \tilde{\pi}) = 0.$$

The residual function $R(x; \tilde{\alpha}, \tilde{\pi})$ is continuously differential by Assumption 5 (full rank). Therefore, the Local Inverse Theorem (Ambrosetti and Prodi (1995)) implies that $(\tilde{\alpha}, \tilde{\pi})$ can be locally calculated if there exists $d = \dim(\alpha) + \dim(\pi)$ distinct points $x_1, \dots, x_d$ such that,

$$\text{Rank} \begin{pmatrix} \partial_{\tilde{\alpha}_0} R(x_1; \tilde{\alpha}, \tilde{\pi}) & \partial_{\tilde{\alpha}_1} R(x_1; \theta_0, \pi) & \dots & \partial_{\tilde{\pi}_{d-2}} R(x_1; \tilde{\alpha}, \tilde{\pi}) \\ \vdots & \vdots & \vdots & \vdots \\ \partial_{\tilde{\alpha}_0} R(x_d; \tilde{\alpha}, \tilde{\pi}) & \partial_{\tilde{\alpha}_1} R(x_d; \theta_0, \pi) & \dots & \partial_{\tilde{\pi}_{d-2}} R(x_d; \tilde{\alpha}, \tilde{\pi}) \end{pmatrix} = d.$$

Lemma 2 of Gao and Wang (2023) shows that this is equivalent to the condition in Assumption 5. Therefore, under Assumptions 4 and 5, if the model is correctly specified then $(\tilde{\alpha}, \tilde{\pi})$ is locally identifiable. $\square$

Lemma 3 (FKG Inequality). *Let (X, ϵ, η) be mutually independent random variables, and let $g(\cdot, \cdot) : \mathbb{R}^2 \mapsto \mathbb{R}$ and $h(\cdot, \cdot) : \mathbb{R}^2 \mapsto \mathbb{R}$ be non-decreasing functions in their first arguments. Then, it follows that $\text{Cov}(g(X, \epsilon), h(X, \eta)) \geq 0$. Similarly, if $g(\cdot, \cdot)$ is non-decreasing and $h(\cdot, \cdot)$ is non-increasing, then $\text{Cov}(g(X, \epsilon), h(X, \eta)) \leq 0$.*

Proof of Lemma 3. First, we show the result for non-decreasing functions $g(\cdot)$ and $h(\cdot)$.

Define $X_1, X_2 \stackrel{\text{iid}}{\sim} X$, $\eta_1, \eta_2 \stackrel{\text{iid}}{\sim} \eta$, and $\epsilon_1, \epsilon_2 \stackrel{\text{iid}}{\sim} \epsilon$. Then notice that using the tower property,

$$\begin{aligned} & \mathbb{E}\{(g(X_1, \epsilon_1) - g(X_2, \epsilon_2))(h(X_1, \eta_1) - h(X_2, \eta_2))\} \\ &= \mathbb{E}\{\mathbb{E}\{(g(X_1, \epsilon_1) - g(X_2, \epsilon_2))(h(X_1, \eta_1) - h(X_2, \eta_2)) \mid X_1 > X_2\}\} \\ &\geq \mathbb{E}\{\mathbb{E}\{(g(X_2, \epsilon_1) - g(X_2, \epsilon_2))(h(X_2, \eta_1) - h(X_2, \eta_2)) \mid X_1 > X_2\}\} \quad (\text{I}) \\ &= 0. \end{aligned}$$

Where (I) follows from the monotonicity inequality, which says if the random variable $Y_1 \geq Y_2$ then $\mathbb{E}\{Y_1\} \geq \mathbb{E}\{Y_2\}$. Noticing that $g(\cdot, \cdot)$ and $h(\cdot, \cdot)$ are non-decreasing in their first argument and letting

$g(X_1, \epsilon_1), h(X_1, \eta_1) = Y_1$ and $g(X_2, \epsilon_2), h(X_2, \eta_2) = Y_2$ gives the result. Next, we write,

$$\begin{aligned}
& \mathbb{E}\{(g(X_1, \epsilon_1) - g(X_2, \epsilon_2))(h(X_1, \eta_1) - h(X_2, \eta_2))\} \\
&= \mathbb{E}\{g(X_1, \epsilon_1)h(X_1, \eta_1)\} - \mathbb{E}\{g(X_1, \epsilon_1)\}\mathbb{E}\{h(X_2, \eta_2)\} - \mathbb{E}\{g(X_2, \epsilon_2)\}\mathbb{E}\{h(X_1, \eta_1)\} + \mathbb{E}\{g(X_2, \epsilon_2)h(X_2, \eta_2)\} \\
&= 2\mathbb{E}\{g(X, \epsilon)h(X, \eta)\} - 2\mathbb{E}\{g(X, \epsilon)\}\mathbb{E}\{h(X, \eta)\} \\
&= 2\text{Cov}(g(X, \epsilon), h(X, \eta)).
\end{aligned}$$

So it follows that,

$$\text{Cov}(g(X, \epsilon), h(X, \eta)) = \frac{1}{2}\mathbb{E}\{(g(X_1, \epsilon_1) - g(X_2, \epsilon_2))(h(X_1, \eta_1) - h(X_2, \eta_2))\} \geq 0$$

When $g(\cdot, \cdot)$ is non-decreasing and $h(\cdot, \cdot)$ is non-increasing in their first arguments, then it follows that $-h(\cdot, \cdot)$ is non-decreasing and we can write,

$$\text{Cov}(g(X, \epsilon), h(X, \eta)) = -\text{Cov}(g(X, \epsilon), -h(X, \eta)) \leq 0.$$

□

Proof of Proposition 2.

We show that $\mathbb{E}\{f_\perp \mid X\} = 0$ implies θ_{IV} is weakly causal. Assume that $\mathbb{E}\{f_\perp \mid X\} = 0$. Using Definition 1 and the law of total covariance, we can write,

$$\begin{aligned}
\theta_{IV} &= \frac{\mathbb{E}\{Y f_\perp\}}{\mathbb{E}\{D f_\perp\}} = \frac{\text{Cov}(Y, f_\perp)}{\text{Cov}(D, f_\perp)} \\
&= \frac{\mathbb{E}\{\text{Cov}(Y, f_\perp \mid X)\} + \text{Cov}(\mathbb{E}\{Y \mid X\}, \mathbb{E}\{f_\perp \mid X\})}{\mathbb{E}\{\text{Cov}(D, f_\perp \mid X)\} + \text{Cov}(\mathbb{E}\{D \mid X\}, \mathbb{E}\{f_\perp \mid X\})} \\
&= \frac{\mathbb{E}\{\text{Cov}(Y, f(X, Z) \mid X)\}}{\mathbb{E}\{\text{Cov}(D, f(X, Z) \mid X)\}}.
\end{aligned}$$

Now, under Assumption 1 (excludability), if $D(x_i, z_i, v_i)$ weakly increases in $f(x_i, z_i)$, it follows that $f(x_i, z_i)$ weakly increases in D . Therefore, we can express $f(x_i, z_i)$ as a non-decreasing function in D , with an additional component that is independent of ϵ when conditioned on X under Assumption 1 (excludability). We can do the same when $D(x_i, z_i, v_i)$ weakly decreases in $f(x_i, z_i)$. Then, consider two cases,

1. $Y(d, x, \epsilon)$ is non-decreasing in d . Then, under Assumption 6 (instrument monotonicity), if $D(x_i, z_i, v_i)$ is weakly-increasing in $f(x_i, z_i)$, then Lemma 3 (FKG) implies that both $\text{Cov}(Y, f(X, Z) \mid X) \geq 0$ and $\text{Cov}(D, f(X, Z) \mid X) \geq 0$, so $\theta_{IV} \geq 0$. If $D(x_i, z_i, v_i)$ is weakly-decreasing in $f(x_i, z_i)$, then both $\text{Cov}(Y, f(X, Z) \mid X) \leq 0$ and $\text{Cov}(D, f(X, Z) \mid X) \leq 0$, so $\theta_{IV} \geq 0$.
2. $Y(d, x, \epsilon)$ is non-increasing in d . Then, under Assumption 6 (instrument monotonicity), if $D(x_i, z_i, v_i)$ is weakly-increasing in $f(x_i, z_i)$, then Lemma 3 (FKG) implies that $\text{Cov}(Y, f(X, Z) \mid X) \leq 0$ and $\text{Cov}(D, f(X, Z) \mid X) \geq 0$, so $\theta_{IV} \leq 0$. If $D(x_i, z_i, v_i)$ is weakly-decreasing in $f(x_i, z_i)$, then $\text{Cov}(Y, f(X, Z) \mid X) \geq 0$ and $\text{Cov}(D, f(X, Z) \mid X) \leq 0$, so $\theta_{IV} \leq 0$.

This shows that θ_{IV} satisfies Definition 2 (weakly causal).

The proof given in Proposition 4 of Blandhol et al. (2022) shows that if $\mathbb{E}\{f_\perp \mid X\} \neq 0$, then weak causality is not satisfied. $\square$

Proof of Corollary 2.

Recall from Definition 1,

$$\theta_{IV} = \frac{\mathbb{E}\{Y f_\perp\}}{\mathbb{E}\{D f_\perp\}} = \frac{\text{Cov}(Y, f_\perp)}{\text{Cov}(D, f_\perp)}.$$

Suppose that $\tilde{f}(X)$ is polynomial in X , or linear in the basis $\Phi(X)$. Lemma 1 then implies that $\text{Cov}(D, f_\perp) = 0$, so θ_{IV} is unidentified.

Therefore suppose that $\tilde{f}(X)$ is nonlinear in X . Then,

$$\begin{aligned} \mathbb{E}\{\tilde{f}(X) \mid X = x\} &= \tilde{f}(X) \\ &\neq X \mathbb{E}\{X^\top X\}^{-1} \mathbb{E}\{X^\top \tilde{f}(X)\}. \end{aligned}$$

So Proposition 2 implies that θ_{IV} is not weakly causal. $\square$